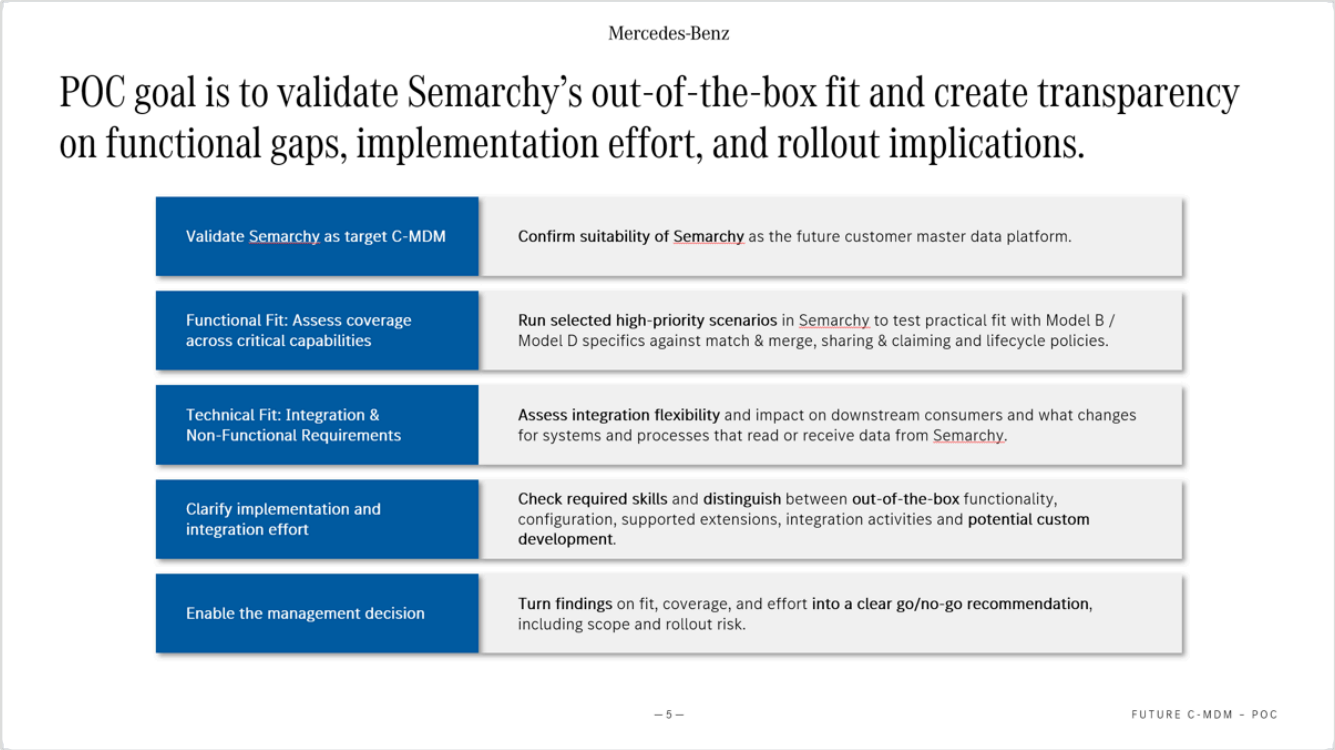
Proof-of-Concept

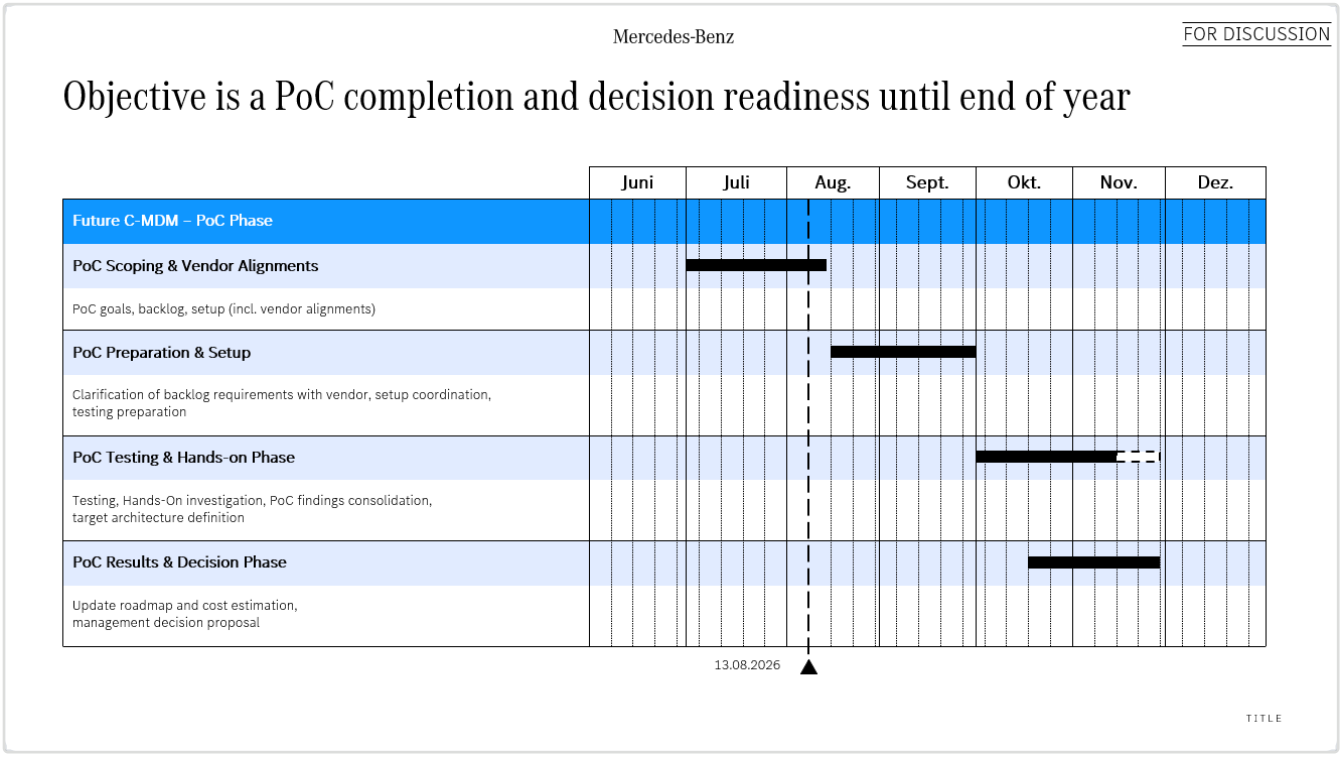


Timeline & Absences

	CW 35 24.08.	CW 36 31.08.-04.09.	CW 37 07.09.-11.09.	CW 38 14.09.-18.09.	CW 39 21.09.-25.09.	CW 40 28.09.-02.10.	CW 41 05.10.-10.	CW 42 12.10.-16.10.	CW 43 19.10.-23.10.	CW 44 26.10.-30.10.	CW 45 02.11.-06.11.	CW 46 09.11.-13.11.
POC Core Topics	***PoC Preparation & Setup*** Focus Topics: 1. Epic 3 - Enterprise policies with retail and data sharing (incl. data model) 2. Epic 1 - Customer Identification and consolidation					***PoC Testing & Hands-On Phase*** Focus Topics: • Epic 2 - End-to-end customer lifecycle • Epic 4 - Technical Fit and NFRs ◦ Eventing with Kafka ◦ Indicative runtime probe						

Absences												
Baldeep												
Yara												
Julian												
Antony												
Holger												
Semarchy, e.g. Derek												
Patrick												
Michael												

Timeline as of 13.08.2026



POC Use Cases

Epic	Journey	Main complexity tested

Epic 1 - Identify & Consolidate ***Daniel*** *Duplicate Journey*	**Operational Data Steward - Customer exists multiple times across dealer, CRM and fleet systems and must be identified, enriched company data, reviewed and consolidated.** 1. Customer and associated company exist multiple times across dealer, CRM and fleet systems 2. Exact and fuzzy matching identify potential B2C and B2B duplicates 3. Ambiguous cases are routed to a data steward for review and decision 4. Records are merged based on survivorship rules, while selected records are protected through do-not-merge decisions 5. Customer relationships are reassigned to the surviving records 6. Wrong merge is discovered and corrected with an un-merge 7. All decisions, changes and source information are retained in the audit trail	• B2B and B2C identification • Exact and fuzzy matching • Human-in-the-loop duplicate resolution • Merge and survivorship processing • Do-not-merge decisions • Relationship reassignment • Traceability and audit
Epic 2 - End-to-end customer lifecycle ***Oscar*** *Omnichannel Journey*	**E2E customer lifecycle - A new private customer is created at a dealer, later registers online, updates her data, purchases a car and then needs to be deleted with pre-deletion dependency checks and audit proof.** 1. Private customer is created at a dealer and identified in real time 2. Customer later registers online and links the MmeID to the existing customer record 3. Customer data from dealer, online and self-service channels are checked (with external address data), enriched (with external company data) and updated. 4. Legal basis for storing customer record is regularly checked (e.g. MmeID/TOAS, DoC, retention periods, ...)	• Real-time identification • Cross-channel account linking • Data validation and enrichment • Controlled self-service updates • Policy enforcement for protected data • Product and customer relationship handling • API and event-based distribution • Data-subject-rights consolidation (incl. DoC)

	5. Customer requests information and deletion about stored personal data and the data is consolidated across connected systems 6. Deletion dependencies and legal-basis is checked 7. Customer data is deleted or anonymised and the deletion result is documented.	• End-to-end (deletion) lifecycle
Epic 3 - Enterprise scale with multi-context & data sharing ***Connie*** *Context Journey*	**Retail and data sharing - Private customer is created at a dealer, later linked to a company and retail contexts, used across markets and legal entities, updated through retail/backchannel APIs and then needs one eligible context to be deleted while remaining relationships are retained.** 1. Private customer is created at a dealer, later linked to a company as an employee or authorized representative 2. Customer is used across different markets, legal entities and B2C, B2B, B2B2C and B2E contexts 3. Shared and context-specific customer data is updated (through retail and backchannel APIs?) 4. Access and update rights are applied based on market, legal entity, role and customer context 5. Relevant customer data is shared across permitted contexts while restricted data remains separated 6. One eligible context is deleted while the remaining customer records, relationships and contexts are retained	• Data model with B2B/B2C/B2B2C/B2E/ ... relationships • Context and legal-entity partitioning and context-specific deletion • Market and legal-entity partitioning • Context-partitioning and update control • Access control with Model D data sharing (claiming)
Epic 4 - Technical Fit & NFRs	**The configured solution is assessed against the key operational and non-functional requirements of Future C-MDM.** 1. Standard APIs are used for representative create, read, update, search and bulk interactions 2. Customer lifecycle changes are distributed through the defined event-driven integration approach 3. Representative read, search and identification scenarios are executed under indicative runtime load	• Operational API integration • Event-driven /Kafka Integration • Performance, latency and throughput • Security and IAM integration

	4. Further important topics for Security, Resilience, Monitoring, etc. are assessed	• Availability and resilience

Backlog

> **2 types of stories within the Epics**
>
> - **Proof** - Deterministic, verifiable outcome. The capability either works or doesn't.
> - **Validation** - Mechanism/concept is demonstrated end-to-end, but full rule depth, coverage, or configuration is not built. Confidence is directional, not exhaustive.

Epic 1 - Identify & Consolidate

Daniel Duplicate Journey

Story	Description and Acceptance Criteria	Responsibility
Provision of test data Type: Precondition	Representative test records available with the following specifics: • *[amount]* of records available ◦ within *[x%]* Examples for 100% matches ◦ within *[x%]* Examples for fuzzy matches ◦ within [x%] examples for clear non-matches (control group, to test false-positive avoidance) ◦ B2B and B2C records both represented • Test records representative for CDM-DIL data quality and structure	CDM-DIL PO or Architect (tbd) with support from Senacor

	• Records in a csv file (to be clarified with Semarchy)	
Exact match de-duplication *Type: Proof* *Effort: Out-of-the-box*	Records that are 100% identical across key attributes are automatically identified as duplicates without manual intervention: • Exact (100%) matches are automatically identified as (confirmed) duplicate. • All 100% duplicates in the test set are identified • No manual step required for exact matches • Clear non-matches are not matched (false positive control group) • Match rules have the following characteristics: ◦ Matching rules can be market specific ◦ To be refined: 100% match is by default defined, if the following field combinations are identical ▪ First Name, Name, E-Mail, Date of Birth ▪ First Name, Name, Phone, Data of Birth ▪ ... • Values for exact match are normalized (abgrenzung fuzzy match)	Configuration: Semarchy - 1h for Match Rule configuration Testing: tbd

Fuzzy match identification *Type: Validation (tbd - exemplary fuzzy rules)* *Effort: Minimal configuration or reuse of examples (Assumption)*	Records with partial similarity are flagged as potential duplicates for further review: • Market specific rules are available • Mechanism validation only, without simulated precision (tbd — confirm whether a precision/recall target is demoed or explicitly out of scope for POC) • Result of fuzzy matches is documented (which fields triggered the match, confidence indicator) ◦ Fuzzy matches above a confidence value (e.g. 90%) are automatically matched as confirmed duplicate. ◦ Fuzzy matches below are marked as ambiguous	Configuration: Semarchy - 4h for Validation Rule and Matching Rule configuration Testing: tbd
Deduplications triggered by external systems *Type: tbd* *Effort:* Low - OOTB REST AP	Other systems need to be able to trigger a deduplication process e.g. via API. Integrity checks need to be run to avoid wrong deduplication (safety guards).	Configuration: Semarchy - 4h for ??
Human-in-the-loop resolution *Type: Validation (tbd - exemplary and simplified task/workflow configuration)* *Effort: Minimal configuration or reuse of examples*	Data steward can see assigned ambiguous matches and decides on the resolution: • Ambiguous/fuzzy matches are routed to a steward for decision	Configuration: Semarchy - 4h for duplicate Suggestion configuration Testing: tbd

(Assumption)	• Steward can approve (merge), reject (no match), or protect (do-not-merge) • Do-not-merge decision is persisted and the record is not proposed again in subsequent match runs	
Merge & survivorship *Type: Validation (tbd - exemplary rule sets, not tuned)* *Effort: Minimal configuration or reuse of examples (Assumption)*	Confirmed duplicates are merged automatically according to pre-configured survivorship rules, without manual field selection • System applies pre-configured survivorship rules automatically on merge ◦ Mechanism validation only, without simulated precision • Surviving record reflects correct field-level winners per rule set • No manual field-by-field selection required (but possible) • Merge protection for certain attribute criteria, e.g. financial account linked	Configuration: Semarchy - 4h for consolidation strategy configuration Testing: tbd
Relationship reassignment *Type: Proof* *Effort: Out-of-the-box*	All relationships held by the losing record in a merge are reassigned: • All relationships (contracts, vehicles, contacts, etc.) are reassigned to the surviving record	Configuration: No configuration required by Semarchy Testing: tbd

	• 0 orphaned relationships post-merge	
Un-merge / correction *Type: Proof* *Effort: Out-of-the-box*	incorrectly executed merge can be reversed: • A completed merge can be reversed • Un-merge restores the original separate records and their respective relationships • Reversal is possible without data loss	Configuration: No configuration required by Semarchy Testing: tbd (EA check?)
Traceability & audit *Type: Proof* *Effort: Out-of-the-box with minimal configuration*	Every automated or manual decision and change made is logged for full accountability • Every merge, un-merge, do-not-merge, and survivorship decision is logged • Log includes who/what/when/source-record-state • 100% of merge/un-merge actions are logged and retrievable	Configuration: Semarchy - 1 h for historization configuration Testing: tbd
Out of Scope	• Production-grade match tuning across all markets and source systems • TBD: Formal benchmarking on large-scale datasets ◦ Alternatives needed, e.g. proven external benchmarks or reference customers • Exhaustive rule catalog for every possible duplicate	

	pattern • End-to-end steward operating model with complex queueing, escalation, and SLA handling • Advanced analytics or dashboarding beyond basic audit/log evidence	

Epic 2 - End-to-end customer lifecycle

Oscar Omnichannel Journey

Story	Description and Acceptance Criteria	Responsibility
Real-time identification *Type: Proof* *Effort: Out-of-the-box Rest API*	When a new customer is created, the system checks in real time whether a matching record already exists: • Response (match found / no match) is returned within 200 ms • 0 duplicate records created for (exact-match customers already present in the test set • If no match is found, a new customer record is created • Match rules from Epic 1 reused	Configuration: Semarchy - 2h for setup via Postman Testing: tbd

Cross-channel account linking *Type: Proof* *Effort: Out-of-the-box* Rest API for Search before create	Customer who later registers online can link their new online account to their existing dealer-created record instead of creating a duplicate • No new record is created, if dealer record already available in the system without an online account ◦ Matching rule: E-Mail/Phone Number + Name is identical • Linked account correctly references the original dealer-created customer ID	Configuration: Semarchy - 4h for setup via Postman Testing: tbd
Data quality check *Type: Validation (tbd - no full rule sets)* *Effort: Minimal configuration or reuse of examples (Assumption)*	Customer data entered through any channel (dealer, online, self-service) is checked against defined rules before being stored. • 100% of records with a missing mandatory field are rejected or flagged, not stored silently • 100% of records with an invalid format (e.g., email, postal code) are rejected or flagged • 100% of rejected/flagged records are visible in the log with the reason for rejection • Validation rule set used in the POC is explicitly documented (which fields, which rules)	Configuration: Semarchy - 4h for Dashboard configuration Testing: tbd

Data Normalization - Phone/Mail *Type: Proof* *Effort: Out-of-the-box*	Customer data is normalized during import • E-Mail/Phone ◦ Semarchy internal functionality	Configuration: Semarchy Testing: tbd
Data Normalization - Address *Type: Proof* *Effort: Implementation* ℹ Changes compared to the initial version	Customer data is normalized during import • Address validation/normalization ◦ Address data is normalized with publicly available HERE API ▪ Question Semarchy: Integration as Rest Enricher? RestClient Enricher | Semarchy Documentation Hub or JavaPluginEnricher | Semarchy Documentation Hub ◦ Demonstrate how custom integration could look like (“happy flow”) ◦ Enriched fields are clearly distinguishable from originally provided fields in the test output (e.g., source flag) ◦ Effort for API integration for custom company data enrichment service is assessed	Configuration: Semarchy - 16h for integration (effort for using test API Provision: MBTI Testing: tbd

Data enrichment - Company Data *Type: Validation (Validate D&B integration; enrichment with other sources out of scope)* *Effort: Out-of-the-box for D&B* ℹ Changes compared to the initial version	Incomplete company data is enriched: • Dun&Bradstreet enrichment is checked with a small data set • Enriched fields are clearly distinguishable from originally provided fields in the test output (e.g., source flag) • Effort for API integration for custom company data enrichment service is assessed	Configuration: Evaluation during Hands-On Phase Testing: tbd
Controlled self-service updates *Type: Proof* *Effort: Rule configuration (Assumption)*	Customers can update their own data through self-service channels within defined boundaries: • Update via API from a mocked customer UI/postman • Update is permitted, if no flag is set for “in critical use” ◦ If flag is set, attribute based rules applied, what is allowed for update (exemplary field based rules, no full rule set). • API shows reason for rejection • Rejected attempts are triggering a task creation for a data steward and are visible in the audit/log	Configuration: Semarchy Testing: tbd

Automated legal foundation check *Type: Proof* *Effort: Middleware needed for external enrichment* ℹ Changes compared to the initial version	Automatic, recurring process checks whether a valid legal foundation for storing the customer's data still exists: • Recurring check runs on a defined schedule (e.g. weekly) • Legal foundation is evaluated against: ◦ Mocked contractual sources (based on TOAS) provided by MBAG team ▪ OAuth ▪ Custom MB system ◦ Question Semarchy: Integration as Rest Enricher? Ⓢ RestClientEnricher | Semarchy Documentation Hub or Ⓢ JavaPluginEnricher | Semarchy Documentation Hub ◦ Field based: Fixed retention period (e.g. 2 years) compared to a field • 100% of test records with no valid legal foundation (absent contract/expired) trigger a deletion request	Configuration: Semarchy - 16 hours API Provision: MBTI `This requires Semarchy DI / ETL since we need to join MDM golden data with external data. It will be easier to join data in SDP Self-hosted since the database is opened for SQL query.` Testing: tbd
Data-subject-rights consolidation *Type: Validation* *Effort: Out-of-the-box for manual export and Rest API.* ℹ	A customer requesting information about their stored personal data receives a consolidated data set: • Export of a complete customer data set can be received	Configuration: Semarchy - 8h (still valid?) `Export via UI into XLSX / CSV. Export via REST API (SDP SaaS) If you want to export into other format and automate the process, this requires Semarchy DI / ETL.` Testing: tbd

Changes compared to the initial version. ◦ manually ◦ REST API ◦ Out-of-scope for PoC (for discussion): automated export		
Deletion with dependency check *Type: Validation (no full rule set and implementation of systems with dependencies)* *Effort: tbd* Changes compared to the initial version. Dependency to completion of story "Automated legal foundation check"	Before a customer is deleted or anonymised, the system checks for active dependencies • 100% of test records with an active dependency/retention flag are blocked from deletion ◦ Reuse data integration of "**Automated legal foundation check**" story • Block reason is displayed/logged in 100% of blocked cases • 100% of test records without dependencies pass the check and proceed	Configuration: Semarchy - 8h API Provision: MBTI `This requires Semarchy DI / ETL since we need to join MDM golden data with data from other systems. It will be easier to join data in SDP Self-hosted since the database is opened for SQL query.` Testing: tbd
Deletion execution and documentation *Type: Proof* *Effort: Out-of-the-box (Assumption)*	Once dependency and retention checks pass, the customer is deleted, archived or anonymised and the outcome is documented. • 100% of cleared test records are deleted, archived or anonymised as configured • Deletion/archived/anonymisation record (what, when, outcome) is created for 100% of executions	Configuration: Semarchy - 8h `Semarchy Data Management can purge records based on retention period. But this requirement additional checks before a customer can be deleted / archived. This requires Semarchy DI / ETL to check across and then submit deletion. It will be easier to fulfil in SDP Self-hosted since the database is opened for SQL query.` Testing: tbd

	• Documented result is retrievable on demand	
Out of Scope	• Integration of channels, source systems and **downstream consumers** ◦ Showcase integration of CRMconnect possible (not in scope by Semarchy) ◦ Showcase integration of downstream systems via Kafka possible (not in scope by Semarchy) • Live integration with all external enrichment or legal-basis providers • Full retention-policy design and recurring compliance operations • Production-grade event topology with all failure, retry, and monitoring patterns	

Epic 3 - Enterprise scale with multi-context & data sharing

Connie Context Journey

Story	**Description and Acceptance Criteria**	**Responsibility**
Multi-context relationship model *Type: Proof (Highest priority)*	Configuration of the target data model based on CBOM 1.2	Configuration: Semarchy - Depends on the Model complexity, it could be 2 to 3 days work

Effort: System configuration (Assumption) ℹ Changes compared to the initial version. Data model available below	• Main CBOM objects and fields are available ◦ Object definition to be defined • 100% of test cases correctly create the intended relationship type (B2C, B2B) • A single customer can hold multiple concurrent context relationships (e.g., B2C and B2E) without one overwriting the other, in 100% of test cases	Testing: tbd
Market and Legal-entity partitioning *Type: Proof (Highest priority)* *Effort: Complex configuration (Assumption)*	Market-dependent customer data separation by legal entity, where required, while approved sharing remains possible through explicit market-level policy. • 100% of test records assigned to Market A are not visible, returned or used in matching when queried from Market B • Market specific policies configured: ◦ Market B: Customer records linked exclusively to Entity A must not be visible, searchable, or returned to users or systems operating in Entity B. ◦ Market D: Customer records can be shared and linked to multiple	Configuration: Semarchy - 4h for Match Rule Binning configuration and Model Privilege Grant configuration Testing: tbd

	entities. Updates on a record affect all Legal Entities. ◦ Without this proof, the record must remain hidden from Entity B.	

Access control and data sharing (claiming)
Type: Proof (Highest priority)
Effort: Complex configuration (Assumption)
ℹ Changes compared to the initial version. No requirement of setting a claim based on pure search interaction.
	An organization (legal entity) can only fully access customer records with a stored and active customer-organization relation and can claim or maintain access only within the defined policy boundaries.
	• A record is visible only if the customer-organization (dealer/legal entity) relation is active
	◦ Note: Visibility =! Searchability
	▪ Visibility = Quick Search
	▪ Searchability = Need to know search
	• Active relation is defined by last interaction + 2 years
	• Each interaction refreshes relation data
	◦ Creation of a duplicate or the attempt via a POST request
	◦ Update of customer data
	◦ ~~Not required: Claim based on successful search of a customer~~
		Configuration: Semarchy - 8h
		`Adding an interaction-date attribute`
		`Populate the interaction-date during Creation / Update`
		`During Search is not possible`
		`Configure Business View filter for displaying active relationship`
		`Create a separate Business View for the “Need to know search” use-case`
		Testing: tbd

	◦ Showcase: API-based update of the relation from an external system • Records without an active relation are not visible, but can be searched. Results only appear if request satisfies the configured proof-of-knowledge rule, requiring at least three approved identification factors ◦ market specific rules, e.g. Market A: Name, E-Mail or VAT Number + Company Name	
Context-specific updates *Type: Proof* *Effort: tbd*	Shared and context-specific customer data can be updated, with updates applied only to the intended context. • Context-specific field affect only the targeted context, not other contexts for the same customer	Configuration: Semarchy - 1h for Model Privilege Grant configuration Testing: tbd
Context-specific deletion *Type: Validation (no full rule set and implementation of systems with dependencies)* *Effort: tbd* Dependency to completion of deletion in Epic 2.	Deletion process (see Epic 2) is evaluated per context and result may differ per context: • Legal foundation is checked independently per context for a multi-context customer • Test cases where one context loses its legal foundation trigger a deletion request for that context only	Configuration: Semarchy - 8h Testing: tbd

	• Other valid contexts are not deleted • Outcome (context, legal basis checked, result) is logged per context, not just per customer	
Out of scope	• All B2B/B2C/B2B2C/B2E combinations beyond the selected test cases • Complete policy framework for every access and sharing scenario • Exhaustive cross-context lifecycle handling for all customer objects • Full organizational operating model for context ownership and approvals	

Epic 4 - Technical Fit and Non-Functional Requirements

Story	Description	Responsibility
Setup Integration Layer Type: Precondition ℹ Changes compared to the initial version.	Install Semarchy xDI • Infrastructure for self-hosted Integration Layer is provided ◦ Infrastructure Requirements: Semarchy xDI System requirements :: Semarchy xDI Documentation • Integration layer is installed and connection to Semarchy SaaS Instance is	Setup: MBTI + Support by Semarchy

	established and initially tested ◦ Tutorial: Install Semarchy xDI on Docker	

API assessment
Type: Validation
Effort: tbd
	Semarchy exposes documented APIs for create, read, update, search, and bulk operations; all required endpoints are callable in the POC and mapped to the target consumer scenarios.
	• **Consumer system integration:** Assessment of standard Semarchy APIs for operational create, read, update and search interactions for customer records
	• **Standard software integration:** Integration patterns for Salesforce, SAP, ... are clarified at architecture and effort level
	• **Legacy and special integration:** Integration of special consumers like CIAM, TOAS, ...
	• **Retailer API/"Rückkanal":** Check use case with "get contacts by change date" (Rückkanal)
	• **Event based intergration:** Clarify distribution (e.g. event based) to the consuming systems for
		Semarchy: Provide API access
		`REST API testing is OOTB SDP SaaS does not support Kafka Publishing SDP Self-hosted support Kafka Publishing: https://docs.semarchy.com/self-hosted/latest/reference/api/dm/data-locations/put-data-notification Recommend this is out-the-scope for SDP SaaS POC.`
		`SDP Self-hosted installation may take days to work with Mercedes infra-team`
		MBTI: Testing

	updates, merge, un-merge, deletion, ... ◦ **Showcase of Kafka publishing:** Exemplary create, update, and delete actions publish Kafka events for 100% of executed test cases within the agreed propagation window • **Showcase:** Connect CRMconnect instance with Semarchy to prove that CRMconnect can connect to Semarchy and complete one end-to-end read/write flow without custom code beyond the agreed integration setup.	
Perform indicative runtime probe *Type: Proof* Effort: To be discussed ℹ Changes compared to the initial version.	Selected runtime-critical interactions are tested indicatively to assess plausibility against operational expectations, e.g. exemplary “load test” for performance and latency under high volume. • Stored records: *xx m* records - tbd • Search and read ◦ Performance / latency for ▪ P75 ≤ 100 ms ▪ P95 ≤ 200 ms ◦ Throughput: ▪ 700 requests per second	MBTI: Testing `Semarchy can provide Mercedes all the API specifications and Mercedes team do the test. Semarchy can provide some internal benchmark for SDP SaaS as a reference..`

	• 500 CMDS + 200 CDM-DIL ◦ Stability: Load for 30-60 minutes • Real-time identification during creation ◦ Performance / Throughput ▪ 140 requests in max. 40 s ▪ Equivalent average throughput: 3.5 requests per second • Stability: Load for 30-60 minutes	
Performance & Latency *Type: Validation (tbd- Proof only with performance testing)* ℹ Changes compared to the initial version.	Access control and policy checks must not make customer lookup or operational processes unusably slow. **Requirements** • Response time < 200 ms for read/query calls, at load of 500/s EMEA and 150/s AMAP (from CMDS; spike is 1000/s) • Real-time identification during creation: 140 requests in max. 40s • policy checks must be executable with low latency • access filtering must scale for high read volumes • performance must remain acceptable even with	MBTI: Testing

	context- and attribute-level restrictions • Data volume: daily requests over the last 30 days is 25,6mio/day (19,1m EMEA, 6,5m AMAP) 95% of that is read (16,1m EMEA, 6,2m AMAP) ▾ Further input Retail Europe: Search and read: P75 100ms Throughput: 100-200 requests per second Real-time identification during creation: 20 requests in max. 40s	
Batch Expectations *Type: Validation*	Initial information: 1 Mio records over night ⚠ Batch NFRs to be delivered by @Dane Holzwarth	
Reference Data Integration *Type: Validation*	Clarify integration of external reference data • GSSN+ • ...	MBTI

Resilience
Type: Validation
	Clarification of Backup, Desaster Recovery, High-Availability
	CMDS:
		`Semarchy SaaS Support Service details : https://semarchy.com/saas-support-policy/`
		`RTO : 24 Hours`
		`RPO : 30 mins Uptime : 99.9%`
		`If Mercedes needs higher SLA can go for SDP Self-hosted`

	6h RTO (Recovery Time Objective) 2h RPO (Recovery Point Objective) Uptime: 98,5 Error Rate average over last 5 min must be below 1%.	

Security
 	
Type: Validation
	Customer master data must be protected against unauthorized access and manipulation.
	
	Requirements
	
	• compliance with internal security policies and standards (e.g. MB security baseline).
	  ◦ Authentication via MB Global Authentication Service (OAuth)
	  ◦ Authorisation via MB Access Management System
	  ◦ To be clarified: Network security (Inbound/Outbound)
	  ▪ IP whitelisting, ...
	• encryption of backup data at rest and in transit
		MBTI: Testing
		
		`Support OAuth`
		`Support Authorization via internal services`
		`Support encryption of data both at rest and in transit`

Data Residency *Type: Validation*	Residency in three regions required • Europe • China ◦ Information Semarchy: Only self-hosted / No	Assumption: Self-hosted for production - no further testing needed

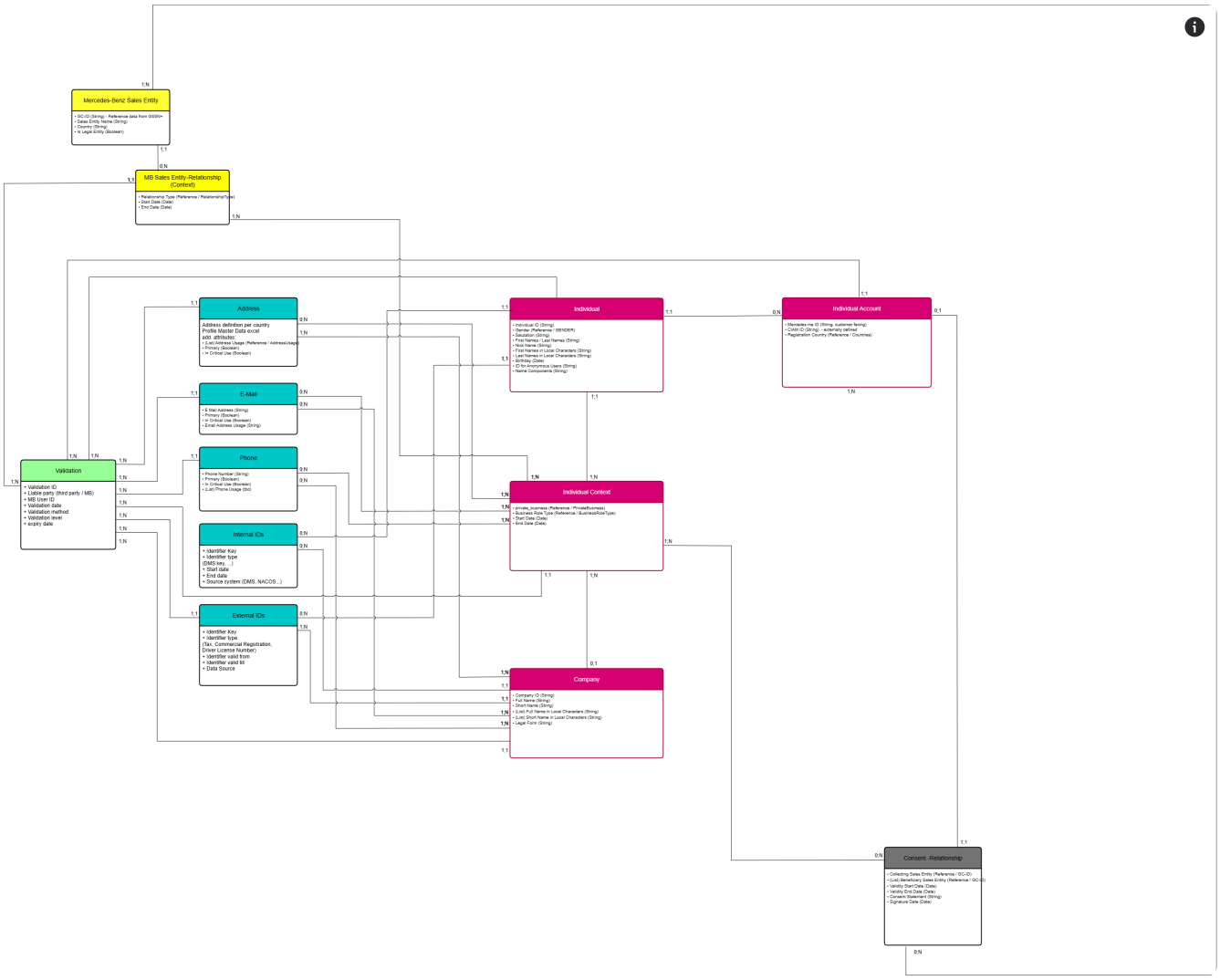
	SaaS • US	
Technical Monitoring *Type: Validation*	Check possibilities for technical monitoring of performance and access for individuals and services.	MBTI
Future Architecture with additional capabilities *Type: Validation*	Assess technical possibilities for integration of external components or systems for customer data related capabilities • Validation • Authentication	

Epic 5 - Implementation effort & decision support

Story	Description	Responsibility
Classify capabilities by delivery type	Each relevant capability is classified as out-of-the-box, configuration, extension, integration or custom effort.	Senacor
Identify migration and rollout drivers	Key migration, transition and rollout drivers are identified for markets, systems, consumers and data domains.	Senacor
Assess license and run-cost implications	Main license, infrastructure and run-cost drivers are identified to support TCO refinement.	Senacor
Prepare management fit/effort matrix	A decision-ready matrix summarizes product coverage, effort, risks and remaining decision points for Option 2.	Senacor

POC Data Model

Simplified based on CBOM 1.2 from LeanIX



CBOM Object	PoC Coverage	Attributes	Cardinality	Comment
`Individual`	In PoC Scope	• Individual ID *(Identifier)* • Gender *(Reference / LoV)* • Salutation *(Reference / LoV)* • First Name(s) *(String)* • Last Name(s) *(String)* • Nick Name *(String)*	• 1:N IndividualContext • 1:N IndividualAccount • 1:N InternalIDs • 1:N ExternalIds • 1:N ConsentRelationship	

		• First Name(s) in Local Characters *(String)* • Last Name(s) in Local Characters *(String)* • Birthday *(Date)* • ID for Anonymous Users *(Identifier)* • Name Components *(String)*	• 1:N MercedesBenzSalesEntityRelationship • 1:N Validation	
`Company`	In PoC Scope	• Company ID *(Identifier)* • Full Name *(String)* • Short Name *(String)* • Full Name in Local Characters *(String)* • Short Name in Local Characters *(String)* • Legal Form *(Reference / LoV)*	• 1:N IndividualContext • 1:N CompanyAccount • 1:N InternalIDs • 1:N ExternalIds • 1:N ConsentRelationship • 1:N MercedesBenzSalesEntityRelationship • 1:N Validation	
`IndividualAccount`	In PoC Scope	• Mercedes me ID *(Identifier)* • CIAM ID *(Identifier)* • Registration Country *(Reference / Country)*	• N:1 Individual	

`IndividualContext`	In PoC Scope	• Context Type (Private / Business) *(Reference / LoV)* • Business Role *(Reference / LoV)* • Start Date *(Date)* • End Date *(Date)*	• N:1 Individual • N:0..1 Company • 1:N Address • 1:N EMail • 1:N Phone • 1:N ConsentRelationship • 1:N MercedesBenzSalesEntityRelationship • 1:N FinancialAccount • 1:N Validation	
`Address`	In PoC Scope	• Address *(Structured Address)* • Address Definition *(Reference / Country-specific)* • Primary *(Boolean)* • In Critical Use *(Boolean)* • Address Usage *(Reference / LoV, multiple)*	• N:1 IndividualContext • 1:N Validation	
`EMail`	In PoC Scope	• Email Address *(String)* • Primary *(Boolean)* • In Critical Use *(Boolean)* • Email Usage *(Reference / LoV)*	• N:1 IndividualContext • 1:N Validation	
`Phone`	In PoC Scope	• Phone Number *(String)* • Primary *(Boolean)* • In Critical Use *(Boolean)* • Phone Usage *(Reference / LoV)*	• N:1 IndividualContext • 1:N Validation	
`InternalIDs`	In PoC Scope	• Identifier Key *(Identifier)*	• N:1 Individual **or** Company	

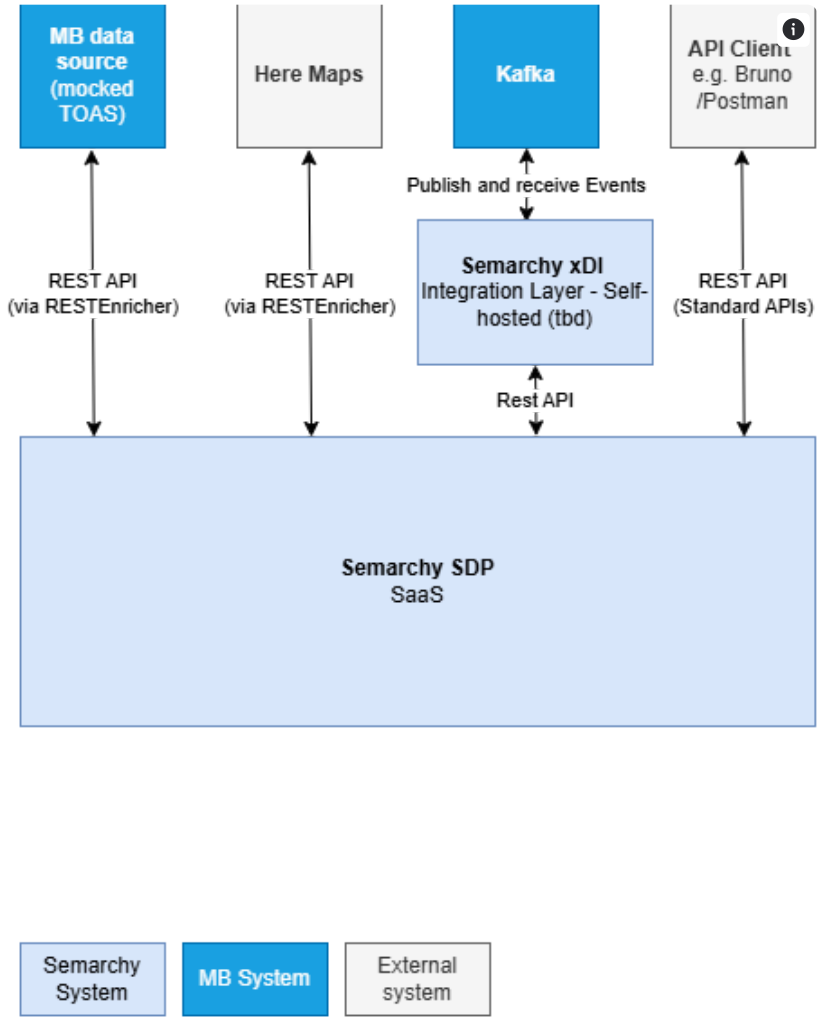
		• Identifier Type *(Reference / LoV)* • Start Date *(Date)* • End Date *(Date)* • Source System *(String or Reference / LoV?)*		
`ExternalIds`	In PoC Scope	• Identifier Key *(Identifier)* • Identifier Type *(Reference / LoV to be defined)* • Valid From *(Date)* • Valid Until *(Date)* • Data Source *(Reference / LoV to be defined)*	• N:1 Individual **or** Company	
`ConsentRelationship`	In PoC Scope	• Collecting Sales Entity *(Reference to* GC-ID*)* • Beneficiary Sales Entity *(Reference, multiple)* • Valid From *(Date)* • Valid Until *(Date)* • Consent Statement *(Reference to* GC-ID*, multiple)* • Signature Date *(DateTime)*	• N:1 Consent • N:1 IndividualContext • N:1 Collecting Sales Entity • N:N Beneficiary Sales Entity	
`MercedesBenzSalesEntity`	In PoC Scope	• GC-ID *(Identifier)* • Sales Entity Name *(String)* • Country *(Reference / Country)* • Legal Entity *(Boolean)*	• 1:N MercedesBenzSalesEntityRelationship • 1:N ConsentRelationship	Prod: Data in sync with MBAG system “GSSN+” PoC: Static reference data, based on export
`MercedesBenzSalesEntityR`	In PoC Scope	• Relationship Type *(Reference / LoV)* • Start Date *(Date)*	• N:1 MercedesBenzSalesEntity	

`elationship`		• End Date *(Date)*	• N:1 IndividualContext	
`Validation`	In PoC scope	• Validation ID *(Identifier)* • Liable Party *(Reference / LoV)* • MB User ID *(Identifier)* • Validation Date *(DateTime)* • Validation Method *(Reference / LoV)* • Validation Level *(Reference / LoV)* • Expiry Date *(Date)*	• N:1 Individual • N:1 Company • N:1 IndividualContext • N:1 Address • N:1 EMail • N:1 Phone • N:1 MercedesBenzUser	
`Consent`	Not in PoC scope	• Consent Name *(String)* • Consent Version *(String)* • Valid From *(Date)* • Valid Until *(Date)* • Contractual Country *(Reference / Country)* • Language *(Reference / LoV)* • Text *(Text)* • Options *(Reference / LoV)*	• 1:N ConsentRelationship	Prod: Multiple external system of records PoC: Static reference data, based on export
`FinancialAccount`	Not in PoC scope	• Financial Account (customer facing) (String) - externally defined • Internal ID (String) • Registration Country *(Reference / Country)* • Registration Country (Reference / Countries)	• N:1 IndividualContext	Simplification fpr POC? External/internal identifier?
`MercedesBenzUser`	Not in PoC scope	• User Account *(Identifier)* • User ID *(Identifier)*	• 1:N Validation • 1:N Sector / Segment • 1:N Customer Group	

`CompanyAccount`	Not in PoC scope	• Company Account Name *(String)* • Company Account ID *(Identifier)* • Registration Country *(Reference / Country)*	• N:1 Company	Not available in as-is systems. Concept feasibility proved by IndividualAccount.
`Membership`	Not in PoC scope	• Membership ID *(Identifier)* • Role *(Reference / LoV)* • VIN *(Identifier, multiple)* • Status *(Reference / LoV)* • Start Date *(Date)* • End Date *(Date)*	• N:1 Customer Group • N:1 IndividualContext	
`CustomerGroups`	Not in PoC scope	• Customer Group ID *(Identifier)* • Customer Group Name *(String)* • Customer Group Responsible *(Reference, multiple)*	• 1:N Membership • N:1 MercedesBenzUser	
`Sector / Segment Participation`	Not in PoC scope	• Start Date *(Date)* • End Date *(Date)* • Status *(Reference / LoV)*	• N:1 Sector / Segment • N:1 IndividualContext	
`Sector / Segment`	Not in PoC scope	• Sector ID *(Identifier)* • Purpose *(Reference / LoV)* • Sector Attributes *(Reference / LoV)* • (List) Responsible *(Reference to* MercedesBenzUser, *multiple)*	• 1:N Sector / Segment Participation • N:1 MercedesBenzUser	

`Fleet`	Not in PoC scope	• Fleet Type *(Reference / LoV)* • Brand *(Reference / LoV)* • Class *(Reference / LoV)* • Size *(Reference / LoV)* • Competitive MB Segment *(Reference / LoV)*	• 1:N Vehicle-Relationship	
`Vehicle-Relationship`	Not in PoC scope	• Individual/Company Vehicle Role *(Reference / LoV)* • Vehicle Usage *(Reference / LoV)* • License Plate *(String)* • Vehicle Acquisition Type *(Reference / LoV)* • In-Car Settings *(Reference / LoV)* • Start Date *(Date)* • End Date *(Date)*	• N:1 Vehicle • N:1 Individual **or** Company • N:1 Fleet	
`Vehicle`	Not in customer master data scope	• Online Code *(Identifier)* • Config ID *(Identifier)* • Order ID *(Identifier)* • Commission ID *(Identifier)* • VIN *(Identifier)* • Brand *(Reference / LoV)* • Vehicle Type *(Reference / LoV)* • Model *(String)* • License Plate *(String)*	• 1:N Vehicle-Relationship	

POC Architecture



Questions for Semarchy 21.08.2026	Result, Open Topics and Follow-Up Needs
Can the PoC Architecture be confirmed?	
Is the PoC Backlog clear enough for you to start the configuration? *Note: Further refinements and clarifications of detail topics possible at any time.*	

Initial configuration with Epic 1 and 3 possible until end of September?	

Archive

Archive with previous initial backlog

#	EPIC	Activities	Addressing Pain Points (exemplary)
0	**PoC foundation & setup**	• Setup environment • Provide test data	-
1	**Functional Fit: Core MDM capabilities** *Focus topic*	• Test customer creation and updates with the different as-is patterns • Validate deduplication against known legacy data-quality issues (existing duplicate volume in source systems)	• Insufficient duplicate prevention during creation • No automated de-duplication of 100% matches
2	**Functional Fit: Policy, governance & lifecycle control** *Focus topic*	• Test exemplary role-, context- and state-based controls for Model D/B specifics • Validate critical-use and validated-data protection • Check deletion governance, auditability and compliance traceability	• High effort for policy changes • Business process limitations, e.g. for data sharing
3	**Functional Fit:**	• Validate a limited set of data quality rules relevant to the core	• Unsufficient/missing quality gate for

	Data accuracy and quality	scenarios • Assess enrichment integration and maintainability • Test manual review and merge clearing together with CDM-DIL/CDM OS stakeholders • Validate data correction and task handling	inbound data • Missing manual duplicate review for (fuzzy) matches • High coordination effort for issue resolution
4	**Customer data model & context handling**	• Assess CBOM and customer model fit • Test B2C/B2B structures and relationships • Validate market contexts, dealer claims and local copies.	• Incomplete representation of customer types • Customer self-service changes not synced • Misuse of additional information fields
5	**Technical Fit: API, access & integration layer**	• Assess APIs and identification flows • Validate integration patterns and runtime behavior	• Complex Interfaces
7	**Implementation effort & decision support** *Core topic*	• Classify standard coverage and configuration effort • Identify extension, integration and custom needs • Prepare the Option 2 management decision view	

PoC foundation & setup

Story	Description	Responsibility
Set up Semarchy PoC environment	Semarchy PoC environment is available, accessible and technically ready for executing the agreed PoC scenarios.	Semarchy
Define measurable PoC evaluation criteria	Evaluation criteria are agreed and can be used measurable and consistently, e.g. for performance and latency.	Senacor
Prepare users, roles and test data	Required PoC users, access profiles and representative test data are available for realistic scenario validation.	tbd
Confirm priority scenarios and pain-point mapping	Priority PoC scenarios are mapped to known business and system pain points to ensure the PoC validates decision-relevant topics.	Senacor

Functional Fit: Core MDM capabilities

Story	Description	Responsibility
Test customer create and update across as-is patterns	Customer creation and update are tested against representative current-state patterns to validate practical operational fit.	tbd

	Including identification before creation for duplicate prevention.	
Validate matching against known legacy duplicate cases	Semarchy matching is tested with representative duplicate examples from legacy systems to assess relevance against known data-quality issues. • 100% matches with automated de-duplication • Fuzzy matching with subsequent manuel review (business usability is validated separately in stewardship epic))	tbd

Functional Fit: Policy, governance & lifecycle control

Story	Description	Responsibility
Configure role- and context-based access	Role- and context-based access behavior can be configured and tested for selected user groups, markets, dealers or legal entities.	tbd
Configure state-based write restrictions	Customer lifecycle states can influence whether data can be changed, restricted or routed for controlled handling.	tbd
Test source-system-specific updates	Update rights can be differentiated by source system or integration context,	tbd

	or the required implementation approach is documented.	
Test critical-use and validated-data protection	Protected customer states and validated attributes can be safeguarded against unauthorized changes in representative scenarios.	tbd
Test deletion governance	Deletion preparation, pre-checks and compliant handling options can be demonstrated or clearly classified by implementation effort.	tbd
Configure merge and survivorship policies	Merge and survivorship logic is configured and can determine golden-record outcomes across selected attributes and sources.	tbd
Assess need for external policy layer	Semarchy's standard governance capabilities are assessed against the target policy needs, with any need for an external policy layer documented.	tbd
Load test with complex policies ⚠ Load test require high-volume test data and could run out of the regular POC capacities by Semarchy	Run a load test with high volume to prove EA requirements with • Response time < 200 ms for read/query calls, • at load of 200 calls per second	tbd

Functional Fit: Data accuracy layer

Story	Description	Responsibility
Configure standard DQ rules	Standard data quality rules are configured in Semarchy and can validate selected customer data attributes.	tbd
Test DQ checks on create/update	Data quality checks can be executed during customer creation or update, with visible validation results.	tbd
Test blocking vs. advisory validation	Semarchy can demonstrate whether validation rules can block transactions, trigger warnings or guide user decisions.	tbd
Test address and company enrichment	Address and company enrichment can be integrated or simulated, with results stored or made available for downstream processing.	tbd
Assess rule maintainability	Data quality rules can be maintained with acceptable effort by the intended operating model, or constraints are documented.	tbd
Configure exemplary steward task handling	Stewardship tasks can be created, assigned, processed and tracked in Semarchy for selected operational scenarios.	tbd

Test manual merge review workflow	Potential duplicate or merge cases can be reviewed manually, decided by a steward and resolved in a traceable way.	tbd
Test data correction workflow	Data quality issues can trigger correction activities that allow users to review, adjust and complete data remediation.	tbd
Test escalation and audit trail	Escalation paths and audit trails for manual decisions are available or the required implementation approach is documented.	tbd
Collect business user feedback	Business users validate whether Semarchy workflows and screens are understandable and usable for future stewardship operations.	tbd

Customer data model & context handling

Story	Description	Responsibility
Assess CBOM alignment and configure B2C and B2B structures	CBOM data model is configured and result is compared against the target CBOM, with deviations and implementation implications documented.	tbd
Configure customer relationships	Relationships between customers, accounts, organizations, dealers and	tbd

	other relevant entities can be modeled and tested.	
Configure market and legal-entity contexts	Market, legal-entity and context-specific customer representations can be configured or otherwise mapped in Semarchy.	tbd